



Snake species classification using deep learning techniques

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Abstract

Incorrect snake identification from the observable visual traits is a major reason of death resulting from snake bites. The classification of snake species has a significant role in determining the appropriate treatment without any delay, the delay may cause dangerous complications or lead to the death of the victim. The difficulty of classifying snakes by human lies in the variations of snake pattern based on geographic variation and age, the intraclass variance is high for specific classes and the interclass variance is low among others, and there may be two remarkably similar types in shape, with one being toxic and the other not. The limitation of the experts' number in the herpetology and their geographical distribution leads us to the importance of using deep learning in the snake species classification. A model to classify snake species accurately is proposed in this study. It is divided into two main processes, detecting the salient object by applying Salient Object Detection (SOD) model based on VGG16 architecture is the first process, the presence of snakes in places with a complex background led to the necessity of separating the salient object, then the classification model is applied with use of image augmentations parameters which improved the results. Four CNN models were used in the classification process including VGG16, ResNet50, MobileNetV2, and DenseNet121. Different experiments on 5, 10, 16, 20, 22, and 45 number of classes and different models were conducted, and the model achieved unprecedented results. The results indicated that the VGG16, DenseNet121, and MobileNetV2 have achieved superior results in the same order from highest to lowest accuracy. The best accuracy is achieved using VGG16 architecture with accuracy 97.09% when using 45 number of classes.

Keywords Salient Object Detection (SOD) · Deep Learning · Snake species classification · VGG16 · DenseNet121

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1 Introduction

Snakes are kind of reptiles that are useful for maintaining ecosystem health. When snakes come across anyone, people usually try to catch or kill them, which can be extremely dangerous or fatal. Since humans do not know the snake species, they consider all the species as poisonous and dangerous reptiles. When Snake species are visually recognized, inaccurate identification may be the result and that could lead to the death of the victim. Identifying snake species is very important to determine the appropriate treatment for snake bites and save human life. The identification can be done manually based on the snake's features such as body texture, head shape, skin color, body shape and eye shape, but the non-expert people are not familiar with these features which causes delay in providing appropriate treatment, leading to the death of the patient [1]. The automatic system based on images has already been developed to classify snake species using the traditional machine learning methods, but in traditional machine learning methods, the applied features and parameters need to be set by an expert in the domain to reduce the data complexity. Nowadays, as a result of the information explosion which has led to huge amounts of data, dealing with large data is very important. But there is also a limitation in the traditional machine learning to deal with large amounts of data. So, the trend is to use what is called deep learning because it has a lot of advantages when dealing with large amounts of data. The most important advantage of deep learning techniques is that they use incremental manner to learn the features. So, the need for an expert in the field will be eliminated. When using deep learning, classification is easier and more accurate. Therefore, deep learning has become increasingly important in image classification due to its ability to extract high-level features from images and learn complex patterns in large datasets. Some popular applications of deep learning in image include semantic segmentation [2, 3], edge detection [4], and object recognition [5].

Üreten, et al. [6] developed a CNN method to help doctors during the diagnosis of rheumatoid arthritis. The dataset which contains 180 radiograph images from the faculty of medicine in Kırıkkale University is used in this study. 81 patients of them are normal and the 99 patients suffer from RA. The dataset is divided into 135 radiographs images for training (74 RA and 61 normal) and 45 for testing (25 RA and 20 normal). The results show that the network achieved 73.33% accuracy, 0.0167 error rate, 0.6818 sensitivity, 0.7826 specificity and 0.7500 precision.

Animal detection using deep learning has become increasingly popular, with applications in areas such as wildlife conservation, veterinary research, and agriculture [7].

Santhanam, et al. [8] proposed a model for detecting animals in the road and give alert to the driver. This model is based on CNN, if the live camera detects an object as an animal, the model gives the driver an alert for 3 seconds. The model achieves accuracy of 91%.

Therefore, the methods of visual recognition of snake species must be replaced by automatic systems where the Classification is based on the features of snakes to be more accurate, thus preserving human life. The systems that were developed in the past were based on traditional machine learning techniques which is a part of Artificial intelligence (AI) [9]. The main examples of the deep learning methods are: Convolutional Neural Networks (CNNs), auto-encoders (AE), Recurrent Neural Networks (RNNs) and Restricted Boltzmann Machines (RBMs). The Convolutional Neural Network is the most popular method in deep learning and consists of three main layers: Convolutional Layer, Pooling layer, and fully connected layer. The general architecture of the CNN is shown in Fig. 1 [10]

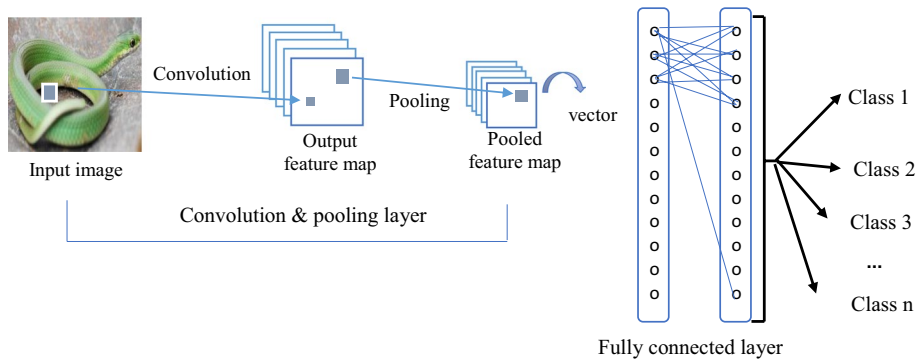


Fig. 1 Convolutional Neural Network architecture [10]

The bite of the snake is one of the deadliest diseases. It causes between 81000 and 138000 deaths and 400,000 victims of malformations and disabilities globally each year. The snake bite affects poor communities in countries which have the highest variety of poisonous snakes. Figure 2 illustrates the mortality of snake bites in the countries of South Asian which are drawn from the community surveys and the hospital records. The Complications occur for about 15% of survivors, these complications include structural malformations, kidney disease, and visual impairment [11].

Delaying treatment, especially after six hours, might be fatal. In certain regions of South Asia, only 50% of patients receive treatment within six hours after being bitten, with the average time from bite to treatment taking 12 to 15 days in some cases. As a result, 70 to 80 percent of fatalities occur before patients get to the hospital. Insufficient treatment results from a lack of experience in health care provider and the absence of epidemiological data on snake bites. There is no sufficient evidence to direct the treatment to the correct protocol. The detection of snake species will help to determine the appropriate treatment which saves humans life and prevent complications [11].

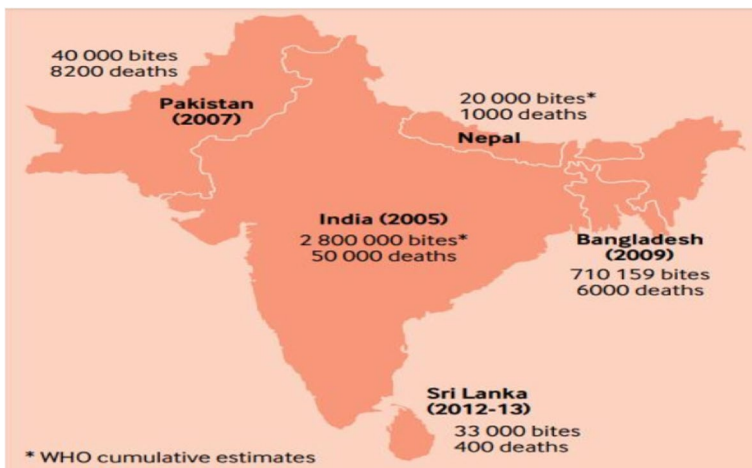


Fig. 2 The mortality of snake bites in the countries of South Asian [11]

Although snake venom poses a threat to human health, it also contains a wealth of bioactive proteins that can be used in drug discovery. For decades, researchers have been studying the evolution of snakes and their venom, mostly by standard morphological and basic genetic approaches, as well as venom proteomics. However, whereas the discipline of genomics has rapidly matured over the last two decades as a result of the advent of next-generation sequencing tools, snake genomics is still in its infancy. According to what is currently known, gene duplication and positive selection are important mechanisms in the neofunctionalization of snake venom proteins. The enormous venom variety and adaptive variation seen in these reptiles can therefore be explained by the fundamental evolutionary drives that snake venom represents. The large number of repeat sequences that have been produced by gene duplication and neofunctionalization in snake genomes also present a significant challenge to DNA sequencing, necessitating the use of significant computational resources and longer sequencing read length for high-quality genome assembly. The detection of snake species will help the chemists to examine the molecular underpinnings of snake venom evolution in unprecedented detail because of continually developing sequencing technologies and computational tools. These groundbreaking discoveries could influence the creation of antivenoms and possibly other medications, as well as advance our understanding of the biology and evolution of snakes [12].

1.1 The list of contributions

This paper presents the following contributions:

- 1- SOD model based on the CNN architecture VGG16 is used to detect the salient object in the image which is the snake.
- 2- Data cleaning is applied to filter the output from the previous step and use only snake's objects.
- 3- An augmentation technique is used as a preprocessing step to the snake's images.
- 4- A classification method using 4 different CNNs architectures is then used to classify a 45 snake species.
- 5- Performance analysis of the model is also presented.
- 6- Comparison between the proposed model and the state-of-the-art.

The remaining of this paper is organized as follows: in Section 2, related work for Snake Species Classification is presented, including the traditional machine learning and deep learning techniques. In Section 3, the proposed model for snake species detection and classification are explained. In Section 4, experimental results of the SOD model are presented and different experiments for the snake classification are conducted using different CNN architectures and different number of classes a discussion of the results is also provided in this section. The conclusion and future work are provided in Section 5 and Section 6, respectively.

2 Related work

Traditional machine learning models often require hand-crafted feature engineering, which can be time-consuming and challenging. In contrast, deep learning models can automatically learn high-level features from raw data, eliminating the need for manual feature

engineering. There are many challenges that faces the researchers, diversity of snake species and the intraclass variance is very high for specific classes, the interclass variance is low between others, the variation in the pattern of the snake based on the snake age and the geographic variation, and There may be two species that are very similar in appearance, one is poisonous, and the other not. So, the snake features analysis is a wide point of interest between medical professionals and computing specialists. All these challenges would lead the researchers to propose model to classify snake species, most of available studies for snake species application use tradition machine learning techniques but the parameters should be manually set. So, researchers direct their efforts to use deep learning models to classify snake species.

Amir, et al. [13] proposed a model for snake species classification, this paper investigates the accuracy of five machine learning techniques: nearest neighbors, k-nearest neighbors (k-NN), decision tree J48, naive Bayes, and backpropagation neural network. The image dataset namely the Snakes of Perlis Corpus consists of 349 images for 22 species that can be exist is Malaysia. William Ratsnake species have the highest number of samples which is 33 samples while Bungarus candidus species have the lowest number of samples which is only three samples. A Color and Edge Directivity Descriptor (CEDD) is used as a texture-based descriptor which uses the characteristics of color and texture in histogram form. Based on texture, Histogram is divided, and six regions is shown. Based on color, 24 regions are calculated from each one of the six regions. That resulted in 144 features. Using JFeaturelib package, the feature vector was generated. The results show that the nearest neighbors and propagation neural network get the best accuracy that 89.22% and 87.93% Respectively. While KNN With K=7 get 80.34%, Naïve Bayes get 75.64% and Decision tree J48 get the lowest accuracy that is 71.29%. Texture-based features only are used but image-based features such as shape and spectral are not applied. The accuracy can be improved using different parameters for KNN and backpropagation neural network. In this paper, only 22 species and 349 images are used, and the accuracy must be improved.

James [14] Proposed a model for classifying snakes' species. For species classification a dataset of 444 images of 6 species were used. The distribution of the dataset 207 images for Spectacled Cobra (SC), 76 for Russel's Viper (RV), 58 for King Cobra (KC), 24 for Common Krait (CK), 36 for Saw Scaled Viper (SSV) and 43 for Hump-nosed Pit Viper (HNPV). For taxonomy analysis, only 88 samples of cobra and 48 samples of viper were used. To identify two different snake species the mean variance filter is used for normalizing image features. The histogram of orientation and gradients is used to create the feature vectors. The total number of features is 31 features. The KNN classifier is used for species classification. After applying the proposed method on the snake feature dataset, the results show that the proposed method gets high accuracy that is 94.27%, and when applying the same method on snake image dataset the results shown that accuracy is low, and it is 85.2%. Only two snake species is classified in this paper, also only 136 images are used in training and 444 images are used in testing.

James, et al. [15] proposed a model to classify snake species. The model is applied to the snake's dataset contains 1299 samples for 6 snake's species, the Spectacled cobra has 490 images, King cobra has 193 images, Common krait has 88 images, Russell's viper has 304 images, saw scaled viper has 116 sample images and Hump Nosed Pit Viper has 108 sample images. For features extraction, 12 Attribute Elevators are involved CfsSubsetEval, FilteredSubsetEval, SymmetricalUncertAttributeEva, ChiSquared AttributeEval, FilteredAttributeEval, InfoGainAttributeEval, ReliefFAttributeEval, OneRAttributeEval, PrincipalComponents, ConsistencySubsetEval, GainRatioAttributeEval, SVMAttributeEval, In addition to mixture of search methods such as Greedy Stepwise, Subset Size

Forward Selection, Genetic Search, Rank Search, Ranker, Linear Forward Selection and Scatter Search. The top 38 features are extracted. The histogram is used to select the most relevant features that have Great influence on the classification. The following 13 classifiers are used for snake's species classification: Ada BoostM1, Naïve Bayes, IB1, Random Sub Space, Bayes Net, Multilayer perception, NB Tree, RBF network, Multi BoostAB, IBk, LWL, J48, and Bagging. The result indicates that only 15 features out of 38 features are enough for classification. When 38 features were used it showed that IBk, IB1 and RBF network algorithms have the best result with the following accuracy: 86.38%, 85.82% and 85.00% respectively. But when only 15 features were used, the following algorithms have the best result: RBF network, IBk and Multilayer perceptron with the following accuracy 88.75%, 87.50% and 86.64% respectively. The accuracy is low, which needs to be improved. Also, only 6 species of snakes were classified, and the dataset contains only 1229 images.

Abdurrazag, et al. [16] applied the classification method for snake species to a dataset contains 415 images for five venomous snake species in Indonesia. The CNN architectures were used, applying convolution operation to a kernel matrix will generate a single activation map, each set of kernel matrices is a local feature extractor. In the pooling layer the process of feature reduction and selection is applied. A pooling algorithm called max pooling is applied to the elements of the activation map to reduce the features by selecting the highest value from neighboring elements. ReLu activation (Rectified Linear Unit) is used to determine which is the best activation function to use based on your network in the Fully connected Layer the class label of the input is determined. When using cross-validation with five-fold, the result shows that the best result is 82% and are achieved by using the medium architecture because the shallow CNN architecture has not sufficient filters while the deep CNN architecture is very complex as it has a lot of filters. But still the number of images in dataset are very few and only 5 snake species is classified also the accuracy needs to be improved.

Abayaratne, et al. [17] proposed a model for classifying snake species. Five CNN models are trained to classify 6 snake species in Sri Lanka. The dataset contains 250 images for each class which were augmented to obtain 2000 images for each class. It is divided into 60%, 20%, 20% for training, validating, and testing, respectively. Four of the used CNN models were applied using the pre-trained architectures ResNet50, VGG16, MobileNet, and InceptionV3 while the remaining model which was used for the classification was developed from scratch. The result indicated that MobileNet architecture achieved the best accuracy of 90.5%. The classes number is very few, also accuracy needs to be improved.

Desingu, et al. [18] proposed a model based on transfer learning technique for snake species classification. The model uses both snake images and their geographic location. The InceptionResNetV2 model is used as a feature extractor, then concatenates the feature vector with geographic feature information. The lightweight gradient boost is used for classification. The used dataset consists of 41442 images belonging to 772 snake classes. It is divided into 347,406 images for training and 38,601 images for testing. The proposed model achieved testing accuracy of 43.2% which is extremely low.

Progga, et al. [19] proposed a CNN-based model to classify snake species into two categories: venomous and non-venomous. The used data set contained 1766 snake images. The transfer learning is used with 7 Neural network models to boost the accuracy of the model. The model achieves accuracy of 91.30% when using cross validation for SGD optimizer with Five-fold. Also, the same model achieves 90.5% without using cross validation. The classification of snake species into venomous and non-venomous aren't sufficient to determine the correct treatment, and the accuracy needs to be improved.

Rajabizadeh, et al. [20] proposed a model for classifying snake species. There are only 6 snake species is used in this study to be classified. Also, few numbers of images are used, only 594 images are used for training and testing. When the LDA and SVM are used, the accuracy was 84%. When using the MobileNetV2 architecture, the model achieves accuracy of 93.16%. This study contains only 6 snake species, and the accuracy must be improved.

Patel, et al. [21] proposed a model, the goal was to classify the racer snakes. The dataset contains 247 images for 9 different classes. There are 4 R-CNN used in this study for image classification, MobileNet, Inception V2, VGG16, and ResNet. The MobileNet, VGG16, Inception V2, and ResNet achieve 10%, 70%, 70%, and 75% respectively. The number of images is very few which represent only 9 classes, also the accuracy is very low.

Yang, et al. [22] presented a model to classify Australian snakes. The used dataset contains 11 snake species from Australia, the dataset contains 1027 images for training and 100 images for testing. For the classification of Australian snakes, several models were used and evaluated such as, YOLOv3, RetinaNet, Tiny YOLO, Faster RCNN ResNet, SSD MobileNet, and SSD VGG16. The results indicate that the Faster RCNN ResNet achieved the best result with Mean Average Performance equal 81.62%. The number of snakes is very few and the result needs to be improved.

Kalinathan, et al. [23] proposed a system for automatic snake classification. This system helps to classify 772 snake species using the SnakeCIEF 2021 dataset which contains 412,537 images. The ResNeXt50-V2 model was used for the classification process and achieved an accuracy of 85.7%. The result must be improved to accurately identify snake species.

Chamidullin, et al. [24] proposed a method for snake species identification and submitted to the SnakeCLEF 2021 challenge by CMP team. The dataset contains 772 snake species and ensemble method based on deep residual neural networks is used for the classification. The model achieved accuracy of 91.6% and F1 score of 0.830.

Vasmatkar, et al. [25] proposed a system based on CNN to classify snake species. The dataset contains 3050 images for 28 snake species, and three CNN architectures including VGG16, MobileNet, and DenseNet were used for the classification process. The results indicated that the accuracy of VGG16, MobileNet, and DenseNet were 12.28%, 58.65%, and 72% respectively. The DenseNet provides the best accuracy, but the accuracy still needs to be improved and the classes number needs to be increased.

Dube, et al. [26] developed a mobile application to classify snake species with aid of the power of CNN. The used dataset in this study were collected from the internet and includes 8 snake species. The images in the dataset were annotated using an online data annotation tool called MakeSense.ai. The YOLOv5 model is used for the training and achieved accuracy of 71%. The model then was converted to Tensorflow Lite to develop a mobile application using java programming language, however the classes number still few and the accuracy needs to be improved.

Dandeniya, et al. [27] presented a web application based on CNN and transformer to classify snake species. An ensemble method is used for 5 CNN models including: EfficientNet B0, EfficientNet B7, ResNet50, ResNet100, Data-Efficient Image Transformers. the model is evaluated in a dataset that contains only 2 classes Russell's Viper and the Indian Rock Python; it contains 500 images for each class so the whole dataset contains only 1000 images. It is divided into 60%, 20%, and 20% for training, validation, and testing respectively. The obtained testing accuracy was 94.5% which is an accuracy that needs to be improved, also the number of images and classes is very few.

Progga, et al. [28] introduced a CNN model to classify snake species, the model consists of 3 components: the first component is feature extraction which uses convolution layer, activation layer, and max-pooling layer to extract features and generate feature maps. The second component is identification of snake which uses a flattened and dense layer to avoid overfitting by reducing the dimensionality of the features. The last component is the classification which uses SoftMax layer to determine the snake species class. The model is evaluated in a dataset comes from Kaggle platform; it contains 1766 images for 2 classes venomous and non-venomous snakes. The accuracy of the proposed model when using five-fold cross-validation for SGD is 91.30%, while the accuracy without cross-validation is 90.50%. The accuracy needs to be improved, the snakes were classified as only venomous and non-venomous snakes, but the snake species weren't detected.

3 Proposed model

The goal of this paper is to develop a model for snake species classification to help health care providers in identifying the snake species to determine the most appropriate treatment as soon as possible and save human life. The model steps are illustrated in Fig. 3

3.1 The proposed technique

The detail of the proposed technique is illustrated in Fig. 4. We first accept images from the dataset as an input to the SOD model to extract the salient object form the image and eliminate any object except snakes. Then, data augmentation techniques are applied including Rescale, shear, zoom, and flipping. The classification process is applied using 4 different CNN architectures to classify snake species. We can say that the system is divided into two main tasks: The first is object detection, the second one is the classification of snake species images. The VGG 16 architecture is used for the process of salient object detection, whereas many CNN architectures are used for the process of image classification.

3.2 Dataset

Most people think that snakes are a single kind of animal, but snakes are very diverse, the biologists are documenting and describing this diversity. Snakes are divided into 24 families, each of which contains 529 genera and 3,789 species. The used dataset is a subset of the snake image dataset which is provided in the snake species identification challenge SnakeCLEF2021. The Conference and Labs of the Evaluation Forum (CLEF) provides



Fig. 3 Proposed Model

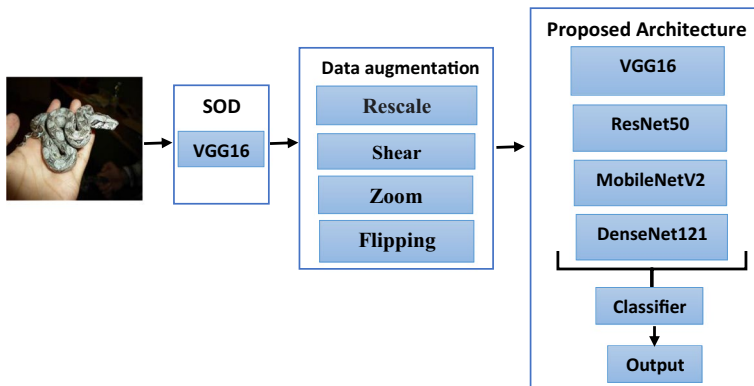


Fig. 4 overview of the snakes' classification Algorithm

data related to biodiversity for the process of identification and prediction. The dataset which consists of 82,601 images belonging to 45 snake species for training is downloaded from Aicrowd website¹. Most of the data were collected from the biodiversity online platforms (such as iNaturalist and HerpMapper). In addition, a total of 17,731 accurately identified images were gathered for testing. The snake classes do not contain the same number of images, the class that has the largest number of images is the *thamnophis_sirtalis* class with 11,092 images, while the class with the smallest number of images is the *lichenura_trivirgata* class with 517 images. Table 1 illustrates the dataset with all 45 snake species and number of images they contain¹. Different snake classes of this dataset had been used in the experiments with total number of classes= 45 classes.

3.3 Data filtering

The first step before starting our work is to apply data filtering. The dataset contains images that can't be opened or images that don't contain snake object. Also, the number of images in the classes aren't balanced. So, we filter the data before starting our work by removing images that cannot be opened or don't contain snakes, select number of images in each class to obtain a balanced dataset. The used dataset contains 11285 images which is divided into 7893 for training and 3392 for testing number of images in each class is illustrated in Table 2.

3.4 Salient object detection (SOD)

The dataset contains images with very complex background. The snakes exist among the rocks, on the trees or in the rivers, also snake may be hold by a human hand and human body is exist in the image, which means that the features of all other objects will be included in the classification of the snakes that will have a bad effect on the performance

¹ <https://www.aicrowd.com/challenges/snakeclef2021-snake-species-identification-challenge#datasets>

Table 1 Snake classes with their number of images in each class

Class id	Original class name	# of images
class 4	thamnophis_proximus	1500
class 67	nerodia_sipedon	3201
class 72	opheodrys_vernalis	852
class 78	crotalus_horridus	2149
class 128	crotalus_pyrrhus	749
class 140	nerodia_rhombifer	1498
class 204	thamnophis_sirtalis	11,092
class 239	natrrix_natrix	677
class 273	crotalus_adamanteus	583
class 326	charina_bottae	639
class 337	pituophis_catenifer	3472
class 362	lampropeltis_triangulum	2154
class 390	nerodia_erythrogaster	2295
class 394	thamnophis_marcianus	1232
class 448	lampropeltis_californiae	2565
class 450	crotalus_ruber	1003
class 460	rhinocheilus_lecontei	1394
class 508	opheodrys_aestivus	2068
class 526	thamnophis_ordinoides	629
class 536	thamnophis_radix	1368
class 540	masticophis_flagellum	1053
class 543	pantherophis_vulpinus	1383
class 561	hierophis_viridiflavus	531
class 581	heterodon_platirhinus	1908
class 597	pantherophis_emoryi	1006
class 629	regina_septemvittata	527
class 635	haldea_striatula	1186
class 639	diadophis_punctatus	2183
class 653	nerodia_fasciata	715
class 654	storeria_occipitomaculata	1071
class 663	crotalus_scutulatus	961
class 697	storeria_dekayi	4619
class 707	crotalus_viridis	1148
class 734	boa_imperator	593
class 771	pantherophis_obsoletus	5536
class 784	lichanura_trivirgata	517
class 804	agkistrodon_contortrix	2555
class 811	thamnophis_elegans	910
class 854	agkistrodon_piscivorus	1773
class 857	pantherophis_guttatus	1138
class 872	crotalus_atrox	5525
class 957	carphophis_amoenus	605
class 966	coluber_constrictor	1471
class 1059	pantherophis_spiloides	890
class 1625	pantherophis_alleganiensis	1677

Table 2 Details of the used dataset with 45 snake species

Class id	Class name	# Of training images	# Of testing images
Class 4	thamnophis_proximus	172	45
Class 67	nerodia_sipedon	387	132
Class 72	opheodrys_vernalis	362	184
Class 78	crotalus_horridus	334	96
Class 128	crotalus_pyrrhus	408	87
Class 140	nerodia_rhombifer	200	89
Class 204	thamnophis_sirtalis	380	132
Class 239	natrix_natrix	154	44
Class 273	crotalus_adamanteus	343	83
Class 326	charina_bottae	151	42
Class 337	pituophis_catenifer	195	96
Class 362	lampropeltis_triangulum	165	56
Class 390	nerodia_erythrogaster	155	49
Class 394	thamnophis_marcianus	158	44
Class 448	lampropeltis_californiae	175	60
Class 450	crotalus_ruber	178	47
Class 460	rhinocheilus_lecontei	165	50
Class 508	opheodrys_aestivus	106	50
Class 526	thamnophis_ordinoides	103	50
Class 536	thamnophis_radix	120	48
Class 540	masticophis_flagellum	110	44
Class 543	pantherophis_vulpinus	146	48
Class 561	hierophis_viridiflavus	102	36
Class 581	heterodon_platirhinos	136	34
Class 597	pantherophis_emoryi	117	60
Class 629	regina_septemvittata	102	39
Class 635	haldea_striatula	123	48
Class 639	diadophis_punctatus	146	60
Class 653	nerodia_fasciata	103	44
Class 654	storeria_occipitomaculata	102	48
Class 663	crotalus_scutulatus	127	48
Class 697	storeria_dekayi	417	132
Class 707	crotalus_viridis	135	44
Class 734	boa_imperator	103	48
Class 771	pantherophis_obsoletus	374	109
Class 784	lichanura_trivirgata	112	48
Class 804	agkistrodon_contortrix	123	48
Class 811	thamnophis_elegans	106	44
Class 854	agkistrodon_piscivorus	110	44
Class 857	pantherophis_guttatus	107	48
Class 872	crotalus_atrox	122	48
Class 957	carphophis_amoenus	108	44
Class 966	coluber_constrictor	114	44
Class 1059	pantherophis_spiloides	113	48
Class 1625	pantherophis_alleganiensis	125	48

of the model. So, the need to apply SOD first appeared and it highly increase the system performance.

SOD is the first step in our mode, we apply a modified VGG16 architecture to our dataset to select the salient object in the image. The SOD algorithm is applied to each class on Google Collaboratory with GPU hardware accelerator. Each class with ≈ 400 images takes approximately 9 hours to save the cropped images for salient objects. Also, there is a need

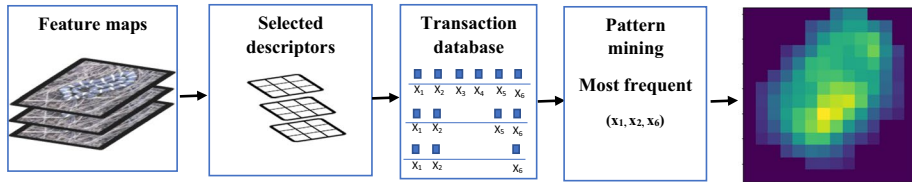


Fig. 5 The detailed process of pattern mining to apply SOD

to another device to run SOD algorithm due to the long time taken by the SOD algorithm for each class on Google Collaboratory, the limited free GPU time given every day, and the need to run the model with a large number of classes as possible.

A device with GPU Nvidia Quadro P1000 with 4G memory, 32G DDR4 and CPU core i7-8850 H is also used parallel to the Google Collaboratory to apply SOD to different classes. But the taken time in this device is approximately double the time spent on Google Collaboratory. It takes approximately 18 hours to run class with ≈ 400 images.

For SOD, the model based on VGG16 architecture which used in [29] is applied. The main idea is to concentrate on unsupervised object discovery when only one image is given. The feature maps from a pre-trained CNN are first converted into a set of transactions to use a pattern mining technique to discover the frequent patterns. The observation of the discovered patterns such as highlighted regions, and co-occurrence hold the consistency of appearance and spatial led the research to merging relevant meaningful patterns to discover and localize objects.

As illustrated in Fig. 5, the snake image is the input as feature maps, then it is fed to the next step which is selected descriptor. The following step is to convert the feature map to a set of transactions, then pattern mining techniques is applied on the feature map to discover the frequent patterns which represent the region of interest.

First, from the pre-trained VGG16 [30] model on ImageNet, (Pool-5 layer and ReLU-5 layer) feature maps are extracted. Also, bilinear interpolation is used to resize feature maps from Pool-5 layer to the same size of ReLU-5 layer.

3.5 Data cleaning

After applying the SOD algorithm and saving the images of salient objects, the importance of the data cleaning step appears. The dataset contains images with very complex backgrounds, also more than one object exists in the image but not all objects are snakes. So, when applying the SOD Algorithm using VGG16, the result contains images for other objects such as hand, Foot, herb, Stone, rock, phone, and so on. So, before we apply the classification algorithm, we must clean the data that are obtained from the previous step by deleting all the unimportant objects.

3.6 Image augmentations

The classification models were trained using image augmentation techniques. It is a method for expanding the training dataset size by creating a modified version of the dataset images. But the image augmentation not only increases the dataset size, but also add a level of variation in the dataset, which helps your model generalize more effectively to data not seen before and solve the problem of model overfitting. Additionally, when the model is trained on new, slightly modified images, it becomes more robust and reliable. The in-place or on-the-fly data augmentation is the type of data augmentation used to implement Keras' ImageDataGenerator class. In this type of data augmentation, when the network is trained, it sees new diversity of our dataset at each epoch. Only images after augmentation are provided to the model instead of using the images in the dataset directly. Because of applying the augmentation randomly, both modified images and almost original images with no augmentation can be used in the training. Because of applying 4 augmentation techniques, the number of images used in each experiment is multiplied by 4 and the data is expanded by creating new versions of the images in the original dataset which will help in the classification process.

Before performing any processing, the input will be multiplied by a value called "rescale." The RGB coefficients in the original images range from 0 to 255, but given a normal learning rate, such values would be too high for the models to handle. As a result, rescale the original images by a factor of $1/255$ is applied which means multiplying the data of the image by the value provided to get images values between 0 and 1 [31].

One of the used image augmentation techniques from Keras' ImageDataGenerator is shear [32] with `shear_range= 0.2`, while the float number represents the shear angle in counterclockwise direction.

The output of applying shear with `shear_range=0.2` is illustrated in Fig. 6.

Also, zoom augmentation [32] is applied to the image. It has two types, zoom out of the image, or zoom in on the image. The class of ImageDataGenerator takes as an argument a float value for `zoom_range`. The zoom will be applied in the range $[1-\text{zoom_range}, 1+\text{zoom_range}]$. A list with two values (the lower and the upper limit) could also be provided instead of using float number. Any value less than 1 will cause the image to zoom in. While any value larger than 1 causes the image to zoom out. The zoom is applied to our dataset with `zoom_range = 0.2`.

The output of applying zoom is `zoom_range = 0.2` is illustrated in Fig. 7.

A flipping augmentation [1] technique is also applied. Flipping images is another good augmentation technique that works well with a variety of different objects. The ImageDataGenerator class has two parameters for flipping along the horizontal or vertical axis: horizontal flip and vertical flip. However, the object in the image should dictate how this technique is used. For example, if an image contains a car object, it would not be sensible to apply vertical flipping of it. Here, a random horizontal flipping is applied.

The output of applying horizontal_flip is illustrated in Fig. 8.

3.7 Classification

The data is finally ready to be classified. During the model training, four model architectures were implemented, VGG16 [30], MobileNetV2 [33], DenseNet121 [34], ResNet50 [35]. Many Experiments were done using these CNN architectures.



Fig. 6 image augmentation using shear



Fig. 7 image augmentation using Zoom

All classification experiments using all models are applied on Google Collaboratory with GPU hardware accelerator parallel with a device with GPU Nvidia Quadro P1000 with 4G memory, 32G DDR4 and CPU core i7-8850 H to apply different experiments to different number of classes using many CNN architectures.

The summary of the specific setup of the experimental environment is shown in Table 3.

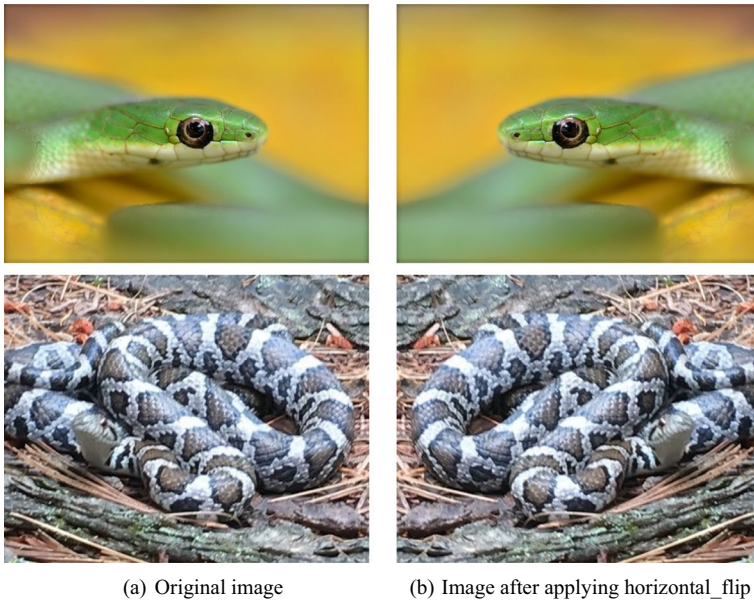


Fig. 8 image augmentation using horizontal_flip

Table 3 Experimental configuration

Configuration items	Parameter values
CPU	Intel(R) Core(TM) i7-8850H CPU @ 2.60GHz 2.59 GHz
Operating system	Ubuntu 16.04 LTS
Other tools	Windows 10 Pro
Operating system	
Memory	32.0 GB
Graphics card	NVIDIA tesla V100
GPU	32 GB
Graphics card	Nvidia Quadro P1000 GPU
Disk	100 GB
Disk	100 GB
Disk	100 GB
Disk	
Other tools	Python 3.6, CUDA 11.2.2

3.7.1 Visual geometry group (VGG) architecture

In 2014, Simonyan and Zisserman [30] proposed an easy and effective design idea for CNN after it was found to be effective in the field of image recognition. Visual Geometry Group (VGG) was the name of this creative design which featured 19 more layers compared to AlexNet. VGG introduced the VGG-16 which contains 138 parameters and used ReLU as activation function. It contains 13 convolutional layers and 4 fully connected layers. The VGG group also introduced VGG-19 which is a deeper version. 224

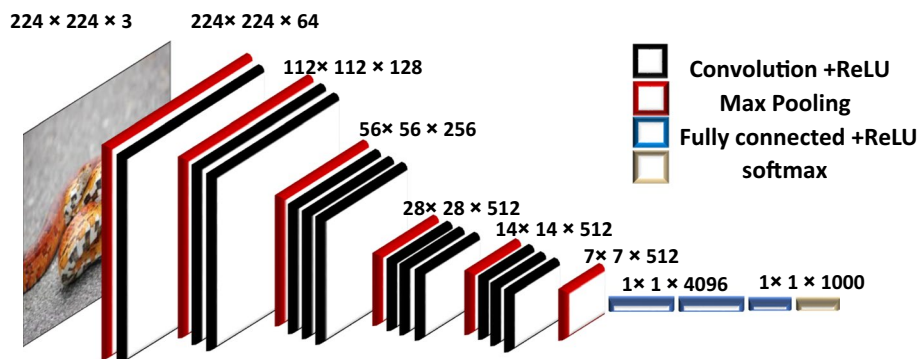


Fig. 9 VGG16 Architecture

× 224 RGB image is the input to VGG. The first two layers contain 64 channels with 3 × 3 filter dimensions and stride 1. A stack of convolutional layers with different depth in different architectures exist After max pooling layer of stride 2. Layers 3 and 4 are convolution layers Which using 128 filters. There are three convolution layers after max pooling with 256 filters. After max pooling layer, there are 6 convolutional layers which use 512 filters with max pooling after each three layers. The following three layers are all fully connected, with the first two layers each having 4096 channels and the third layer having 1000 channels. The final layer is called SoftMax. The architecture of VGG16 is illustrated in Fig. 9.

3.7.2 ResNet-50

The residual network (ResNet-50) was the winner of the 2015 ILSVC challenge and created by He, et al. [35]. It contains 26 million parameters and 50 layers, each ResNet block has two or three convolutional layers. The input image to the network has height and width as multiples of 32. In the first layer of ResNet architecture, the convolution uses 7 × 7 kernel sizes and 64 different kernels, followed by max-pooling using 3 × 3 kernel sizes and stride size of 2. Afterward, there are 3 Residual blocks with 3 layers each. The kernel size in all 3 layers is 64, 64 and 128 respectively and the convolution operation in each Residual block is applied with stride 2. Finally, there are average pooling layers followed by fully connected layers that have 1000 nodes and softmax function at the end. The architecture of ResNet-50 is illustrated in Fig. 10, where the number at the bottom of blocks represents the number of repeating units for each block.

3.7.3 MobileNetV2

In [2], MobileNet V2 model consists of 53 convolution layers and 1 average pooling layer with nearly 350 GFLOP. There are 2 main components, Inverted Residual Block and Bottleneck Residual Block. The MobileNet V2 also contains two types of Convolution layers, 1 × 1 Convolution and 3 × 3 Depthwise Convolution. There are 3 different layers in each block, 1 × 1 Convolution and Relu6, Depthwise Convolution, 1 × 1 Convolution with no linearity. There are Stride 1 and Stride 2 Blocks, Strid 1 block consists of

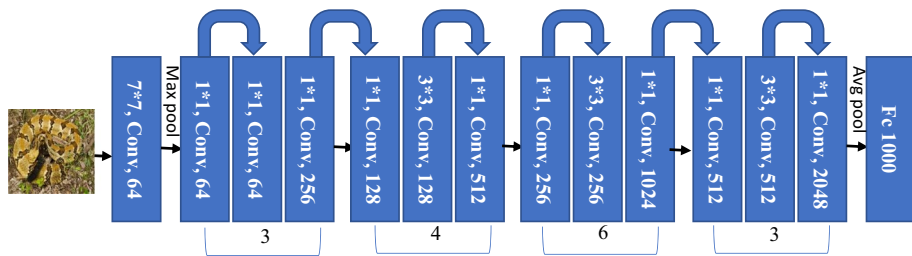


Fig. 10 The architecture of ResNet-50

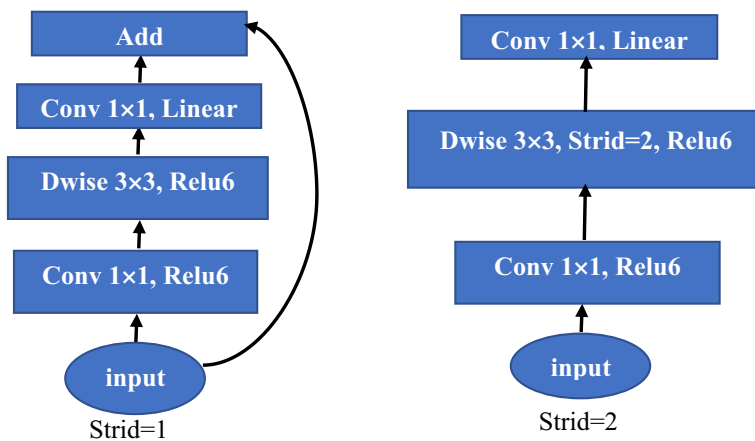


Fig. 11 MobileNet V2 components

input, 1x1 Convolution with Relu6, Depthwise Convolution with Relu6, 1x1 Convolution without any linearity, and Add. Strid 2 block consist of input, 1x1 Convolution and Relu6, Depthwise Convolution with stride=2 and Relu6, 1x1 Convolution with no linearity. The architecture of MobileNetV2 is illustrated in Fig. 11.

3.7.4 Densely connected convolutional networks (DenseNet)

The DenseNet architecture is a modifying version of the standard CNN architecture, The reason for its name is that each layer in DenseNet is connected to all other layers. The number of direct connections is calculated as in Eq. 1, where D represent number of direct and L represents number of layers [34].

$$D = \frac{L(L+1)}{2} \quad (1)$$

The network is divided into multiple dense blocks connected together which facilitate the down sampling process and feature concatenation. The size of the feature map inside dense blocks remains the same. The architecture of DenseNet Fig. 12

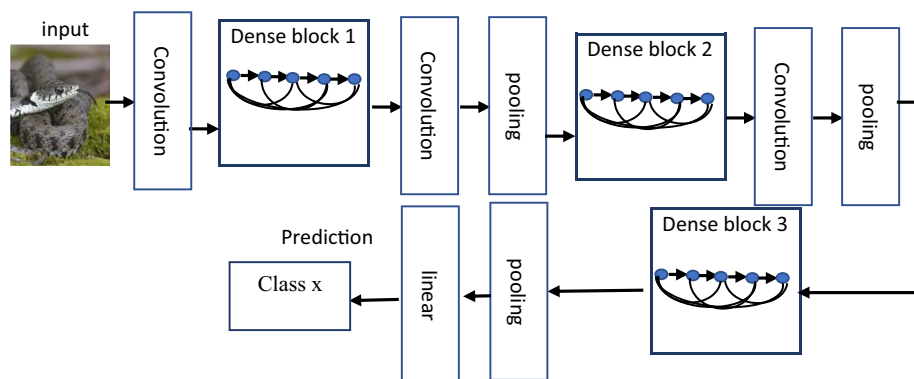


Fig. 12 DenseNet Architecture

3.7.5 Parameters

All the parameters used in the model are discussed, including the activation function, loss function, optimizer, and metrics.

Activation function Softmax is used as an activation function. It is used to calculate the probability distribution and produces the output as a value between 0 and 1, with total probabilities equal to 1. A vector of j real values is transformed into a probability distribution with j potential outcomes using the Softmax function, also known as softargmax or the normalized exponential function. There are 3 possible input values: positive, negative, zero. However, in order to be understood as probabilities, the Softmax function converts them into values between 0 and 1. Softmax is frequently implemented as a layer right before the output layer in neural networks. It needs to have exactly as many nodes as the output layer. the softmax function is calculated as in Eq. 2 [36]

$$F(s_i) = \frac{e^{(s_i)}}{\sum_j e^{(s_j)}} \quad (2)$$

Where s is input vector.

The softmax is used in models that contain multi-class where it calculates the probabilities of each class and target class is with the highest probability. It is used in almost all deep learning architectures in the output layers.

Loss function The loss function is used to judge the effectiveness of the network. In this study, the Categorical Cross-Entropy loss is used, it is also known as softmax Cross-Entropy loss. It is used for multi-class classification with two or more output labels. When actual-value labels are one-hot encoded, which means that only one piece of the encoded data is true at a time, this technique is used. The mean cross-entropy for the entire training dataset is used to compute the overall model's classification error. Using the prediction's negative natural logarithm, the loss for the hot class is calculated the Categorical Cross-Entropy loss is calculated as in Eq. 3 [37]

$$CE = -\log \left(\frac{e^{s_p}}{\sum_j^c e^{s_j}} \right) \quad (3)$$

Where s_p represents the CNN score for the positive classes.

Optimizer Adam [38] optimizer is the used algorithm for optimization, the name Adam is derived from adaptive moment estimation. the adam is applied to update the weights of the network perfectly in an iterative manner in the training data. Adam uses estimation of first and second gradient descent to adapt the parameter of learning rate for each weight in the NN. The first moment is mean while the second moment is uncentered variance (don't subtract mean during variance calculation). The learning rate is the proportion that weights are updated, the default learning rate is 0.001 is used. Larger values lead to faster learning before the rate is updated while smaller values slow the learning rate during the training. The default value for beta1 in Adam is 0.9 which represents the exponential decay rate for the first moment estimates, while the default value for beta 2 is 0.999 which represents the exponential decay rate for the second-moment estimates, the value of beta 2 should be close to 1.0. Equations 4 and 5 show, respectively, the estimation of the first and second moments of the gradients.

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \quad (4)$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \quad (5)$$

Where m_t represents the estimation of the first moments of the gradients and v_t represent estimates of the second moments of the gradients. The exponential decay rates for these moment estimates are represented by β_1 and β_2 . At time step t , g_t denotes the gradient of the loss function with respect to the weights. The squared gradient g_t^2 is used to update the gradients' second moment estimate v_t .

The bias correction is used to compensate for the first and second moment estimates' initialization bias towards zero. Equations 6 and 7 represent the correction of the first and second moment, respectively.

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t} \quad (6)$$

$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t} \quad (7)$$

The update rule for the i_{th} weight at time step t in the Adam optimization algorithm changes the weight $\theta_{i,t}$ to $\theta_{i,t+1}$. Equation 8 used to update the weights iteratively. The α represents the learning rate and ϵ represents small constant added for numerical stability.

$$\theta_{i,t+1} = \theta_{i,t} - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \quad (8)$$

Evaluation metrics The evaluation of the models is based on accuracy metrics, Table 4 illustrates the confusion matrix which will be used to calculate the accuracy of the model.

Accuracy is calculated as shown in Eq. 9

Table 4 The confusion matrix

		Predicted class	
		Yes	No
Actual Class	Yes	True Positive (TP)	False negative (FN)
	No	False Positive (FP)	True negative (TN)

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} * 100 \quad (9)$$

4 Experimental results and discussion

The proposed model is divided into 2 main processes 1) SOD 2) Classification. The first step is to apply SOD.

4.1 Salient object detection (SOD)

The output of applying the SOD algorithm to 4 different classes are shown in Fig. 13 where (a) represents the input image to the SOD algorithm from different classes, (b) are the output images with red box on the region of the salient object, and (c) represents the region of interest or the center points.

4.2 Snake species classification

Different experiments are applied to different number of classes using different CNN architectures. In the 1st, 2nd, 3rd, 4th, 5th, and 6th experiments 5, 10, 16, 20, 22, and 45 snake species are included, respectively.

4.2.1 CNN-based 5 snake species

The distinct CNN models are applied on only 5 snake species, the class-id of the snakes that are included in this experiment are Class 67, Class 204, Class 337, Class 697, and Class 771. Also, different epochs are used with batch size = 64. Total number of images= 2354, training images = 1753 images, and testing images= 601 images. Samples of training and testing images in class 67 is shown in Fig. 14. Also, Samples of training and testing images in class 204 are shown in Fig. 15.

All applied models are tested using data not seen before that was not involved in the training process.

The accuracy of CNN-based 5 snake species are shown in Table 5, and the plot diagram when using 20 epochs and VGG16 architecture is shown Fig. 16, plot between train accuracy and validation accuracy is shown in Fig. 16(a), plot between train loss and validation loss is shown in Fig. 16(b).

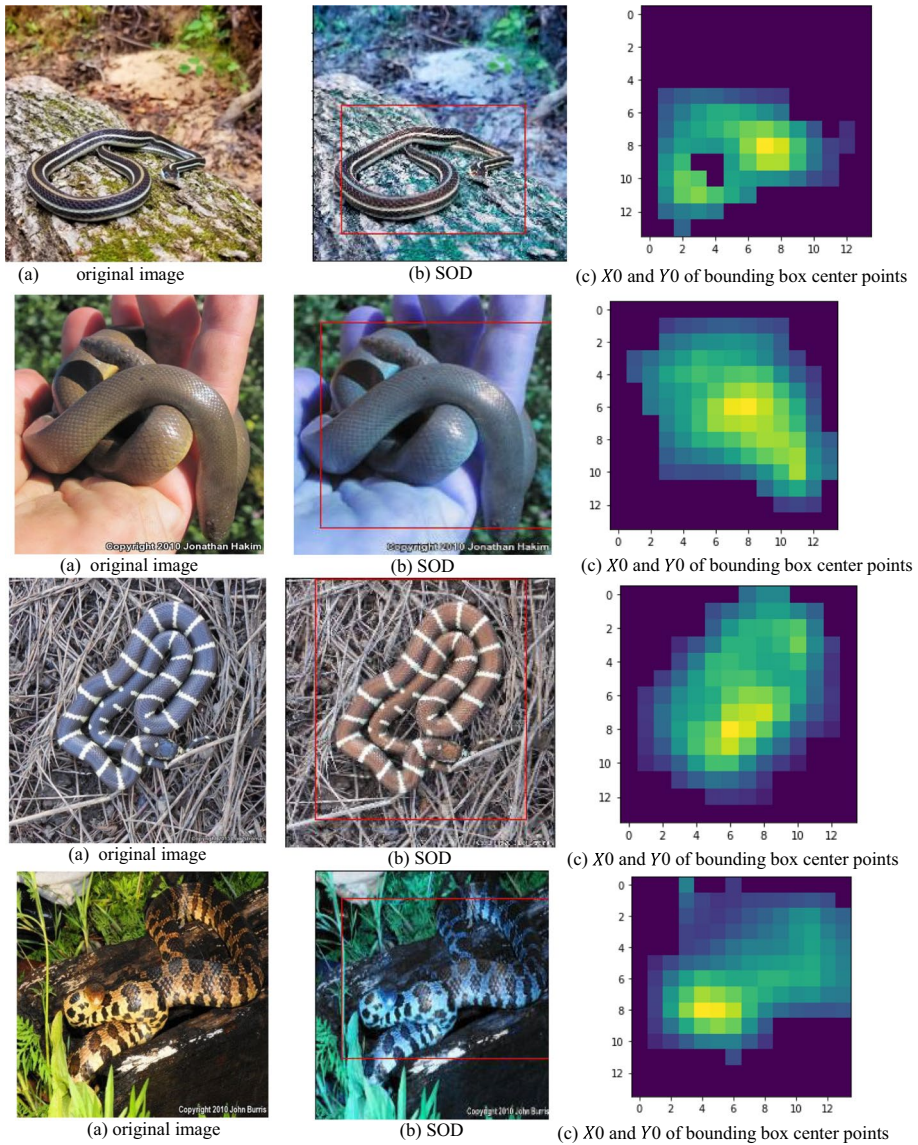


Fig. 13 SOD for different classes

The plot diagram when using 35 epochs and VGG16 architecture is shown Fig. 17, plot between train accuracy and validation accuracy is shown in Fig. 17(a), plot between train loss and validation loss is shown in Fig. 17(b).

The plot diagram when using 35 epochs and ResNet50 architecture is shown Fig. 18, plot between train accuracy and validation accuracy is shown in Fig. 18(a), plot between train loss and validation loss is shown in Fig. 18(b).



(a) Training samples

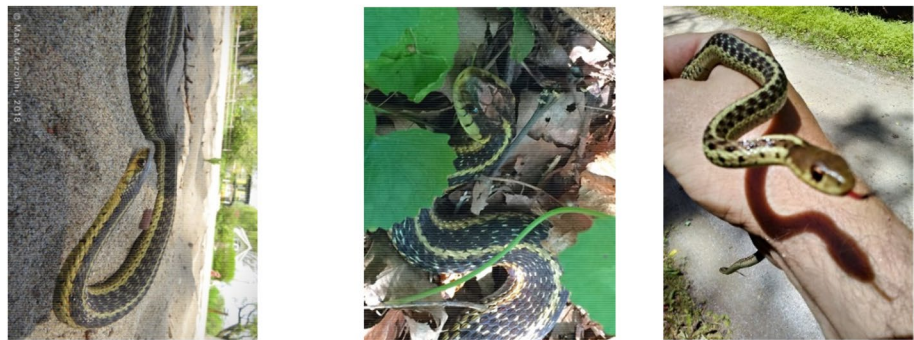


(b) Testing Samples

Fig. 14 class 67 samples



(a) Training samples

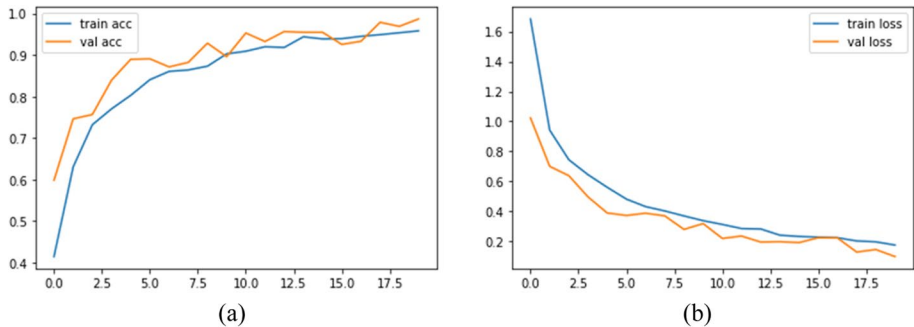
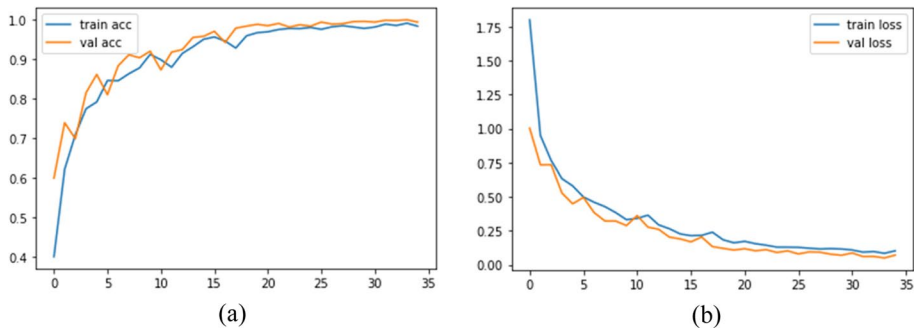


(b) Testing samples

Fig. 15 class 204 samples

Table 5 CNN-based 5 snake species results

Model	# Of Epochs	Accuracy	Batch size	Time for each epoch
VGG16	20	95.3%	64	≈ 40 s
VGG16	35	99.32%	64	≈ 40 s
ResNet50	35	53.79%	64	≈ 40 s
MobileNetV2	35	95.09%	64	≈ 30 s
DenseNet121	35	98.92%	64	≈ 43 s

**Fig. 16** Results plot of VGG16 with 20 epochs& 5 classes. X-axis represents the epoch and y-axis represents the accuracy in (a) and loss in (b)**Fig. 17** Results plot of VGG16 with 35 epochs& 5 classes. X-axis represents the epoch and y-axis represents the accuracy in (a) and loss in (b)

The plot diagram when using 35 epochs and MobileNetV2 architecture is shown in Fig. 19, plot between train accuracy and validation accuracy is shown in Fig. 19(a), plot between train loss and validation loss is shown in Fig. 19(b).

The plot diagram when using 35 epochs and DenseNet121 architecture is shown in Fig. 20, plot between train accuracy and validation accuracy is shown in Fig. 20(a), plot between train loss and validation loss is shown in Fig. 20(b).

As Shown in Table 5, the results indicates that the VGG16 architecture achieves the best performance with accuracy = 99.32% when applying to 5 classes with epochs =35.

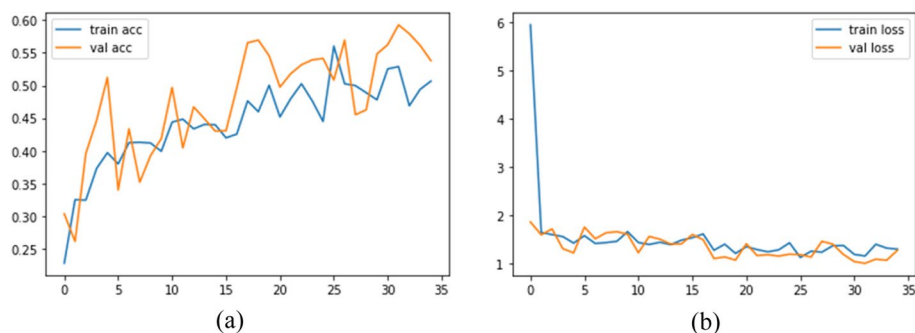


Fig. 18 Results plot of ResNet50 with 35 epochs& 5 classes. X-axis represents the epoch and y-axis represents the accuracy in (a) and loss in (b)

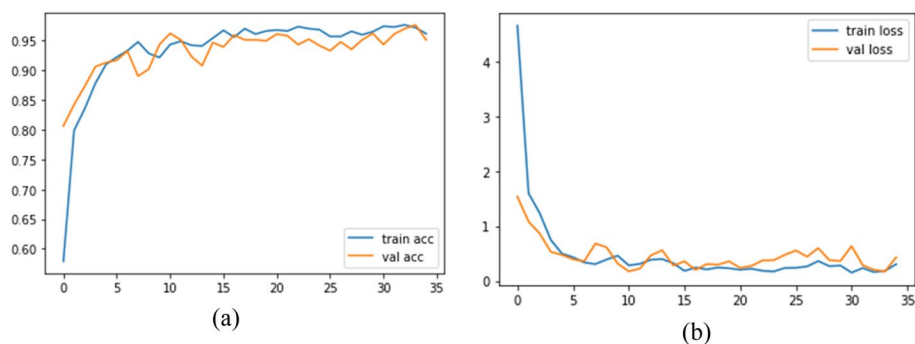


Fig. 19 Results plot of MobileNetV2 with 35 epochs& 5 classes. X-axis represents the epoch and y-axis represents the accuracy in (a) and loss in (b)

4.2.2 CNN-based 10 snake species

The distinct CNN models are applied on 10 snake species. The class-id of the snakes that are included in this experiment are Class 67, Class 204, Class 337, Class 697, Class 771, Class 4, Class 72, Class 78, Class 128, and Class 140. Also, 35 epochs are used with batch size = 64. Total number of images = 4331, training images = 3229 images, testing images = 1102 images. Samples of training and testing images in class 4 are shown in Fig. 21.

All applied models are tested using data not seen before that was not involved in the training process.

The accuracy of CNN-based 10 snake species is shown in Table 6, and the plot diagram when using 35 epochs and DenseNet121 architecture is shown in Fig. 22, plot between train accuracy and validation accuracy is shown in Fig. 22(a), plot between train loss and validation loss is shown in Fig. 22(b).

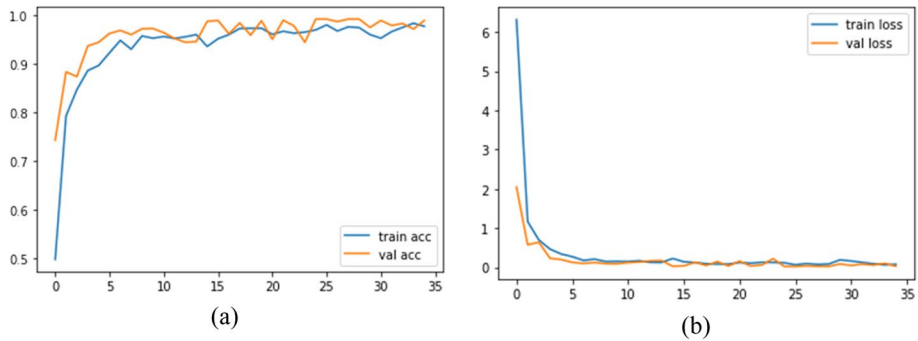


Fig. 20 Results plot of DenseNet121 with 35 epochs & 5 classes. X-axis represents the epoch and y-axis represents the accuracy in (a) and loss in (b)



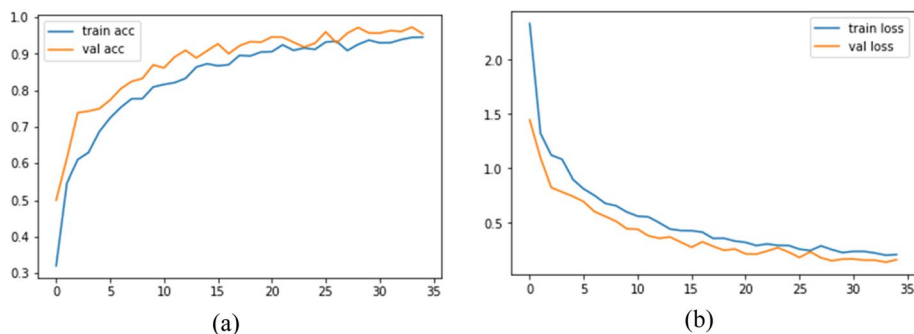
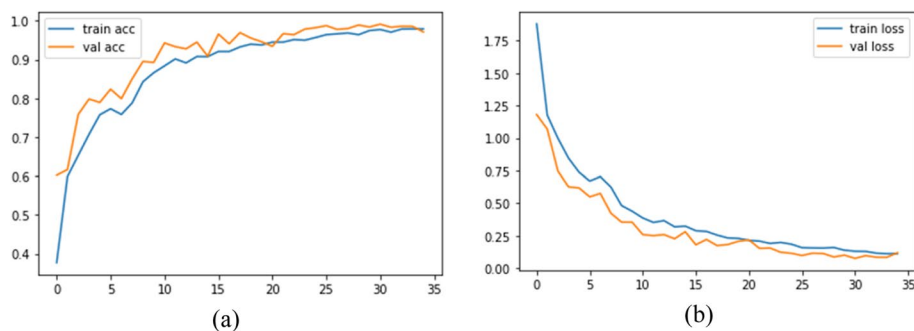
Fig. 21 class 4 samples

The plot diagram when using 35 epochs and VGG16 architecture is shown in Fig. 23, plot between train accuracy and validation accuracy is shown in Fig. 23(a), plot between train loss and validation loss is shown in Fig. 23(b).

The plot diagram when using 35 epochs and MobileNetV2 architecture is shown in Fig. 24, plot between train accuracy and validation accuracy is shown in Fig. 24(a), plot between train loss and validation loss is shown in Fig. 24(b).

Table 6 CNN-based 10 snake species results

Model	# Of Epochs	Accuracy	Batch size	Time for each epoch
DenseNet121	35	95.45%	64	≈ 100 s
VGG16	35	97.12%	64	≈ 90 s
MobileNetV2	35	96.78%	64	≈ 50 s

**Fig. 22** Results plot of DenseNet121 with 35 epochs& 10 classes. X-axis represents the epoch and y-axis represents the accuracy in (a) and loss in (b)**Fig. 23** Results plot of VGG16 with 35 epochs& 10 classes. X-axis represents the epoch and y-axis represents the accuracy in (a) and loss in (b)

4.2.3 CNN-based 16 snake species

The distinct CNN models are applied on 16 snake species, the class-id of the snakes that are included in this experiment are Class 67, Class 204, Class 337, Class 697, Class 771, Class 4, Class 72, Class 78, Class 128, Class 140, Class 239, Class 273, Class 326, Class 362, Class 390, and Class 394. Also, 35 epochs are used with batch size = 64. Total number of images= 5775, training images= 4355 images, and testing images= 1420 images. Samples of training and testing images in class 326 are shown in Fig. 25.

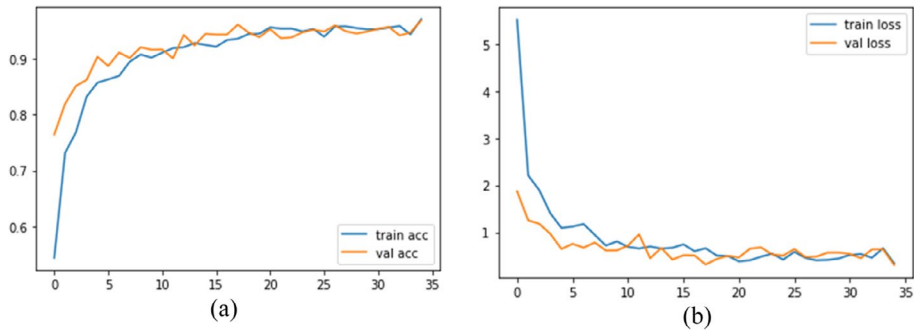


Fig. 24 Results plot of MobileNetV2 with 35 epochs& 10 classes. X-axis represents the epoch and y-axis represents the accuracy in (a) and loss in (b)

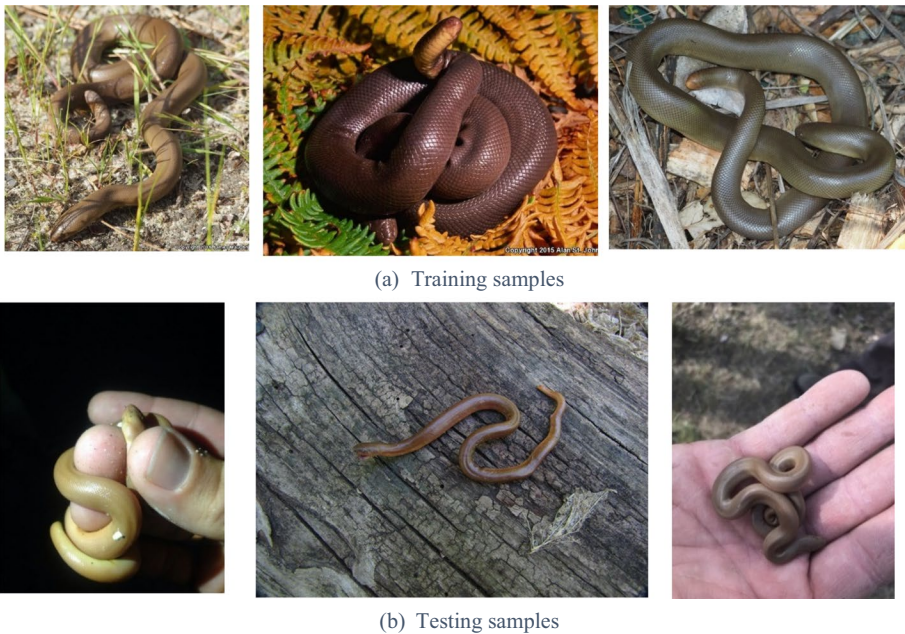


Fig. 25 class 326 samples

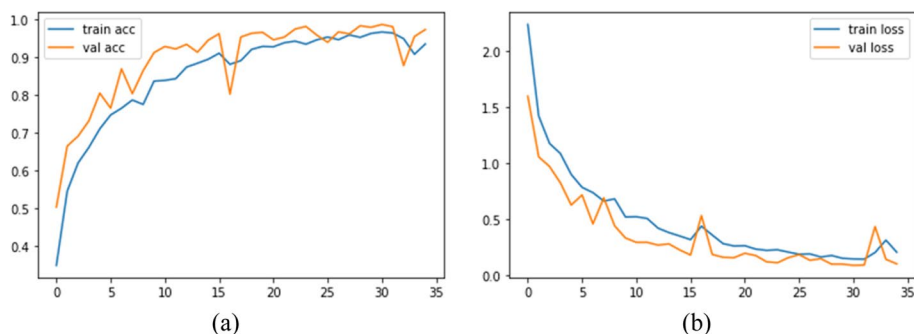
All applied models are tested using data not seen before that was not involved in the training process.

The accuracy of CNN-based 16 snake species is shown in Table 7, and the plot diagram when using 35 epochs and VGG16 architecture is shown in Fig. 26, plot between train accuracy and validation accuracy is shown in Fig. 26(a), plot between train loss and validation loss is shown in Fig. 26(b).

The plot diagram when using 50 epochs and VGG16 architecture is shown in Fig. 27, plot between train accuracy and validation accuracy is shown in Fig. 27(a), plot between train loss and validation loss is shown in Fig. 27(b).

Table 7 CNN-based 16 snake species results

Model	# Of Epochs	Accuracy	Batch size	Time for each epoch
VGG16	35	97.38%	64	≈ 125 s
VGG16	50	98.53%	64	≈ 125 s
DenseNet121	35	94.83%	64	≈ 200 s
DenseNet121	50	96.97%	64	≈ 110 s

**Fig. 26** Results plot of VGG16 with 35 epochs & 16 classes. X-axis represents the epoch and y-axis represents the accuracy in (a) and loss in (b)

The plot diagram when using 35 epochs and DenseNet121 architecture is shown in Fig. 28, plot between train accuracy and validation accuracy is shown in Fig. 28(a), plot between train loss and validation loss is shown in Fig. 28(b).

The plot diagram when using 50 epochs and DenseNet121 architecture is shown in Fig. 29, plot between train accuracy and validation accuracy is shown in Fig. 29(a), plot between train loss and validation loss is shown in Fig. 29(b).

4.2.4 CNN-based 20 snake species

The distinct CNN models are applied on 20 snake species, the class-id of the snakes that are included in this experiment are Class 67, Class 204, Class 337, Class 697, Class 771, Class 4, Class 72, Class 78, Class 128, Class 140, Class 239, Class 273, Class 326, Class 362, Class 390, Class 394, Class 448, Class 450, Class 460, and Class 543. Also, different epochs are used with batch size = 64. Total number of images= 7245, training images= 5018 images and testing images= 2227 images. Samples of training and testing images in class 448 are shown in Fig. 30.

All applied models are tested using data not seen before that was not involved in the training process.

The accuracy of CNN-based 20 snake species is shown in Table 8, and the plot diagram when using 35 epochs and VGG16 architecture is shown in Fig. 31, plot between train accuracy and validation accuracy is shown in Fig. 31(a), plot between train loss and validation loss is shown in Fig. 31(b).

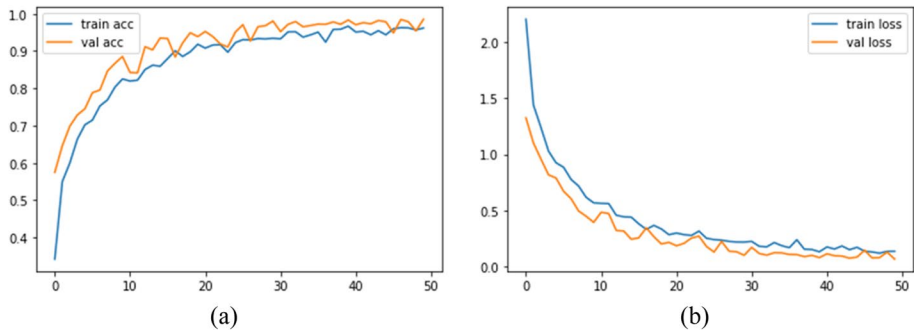


Fig. 27 Results plot of VGG16 with 50 epochs & 16 classes. X-axis represents the epoch and y-axis represents the accuracy in (a) and loss in (b)

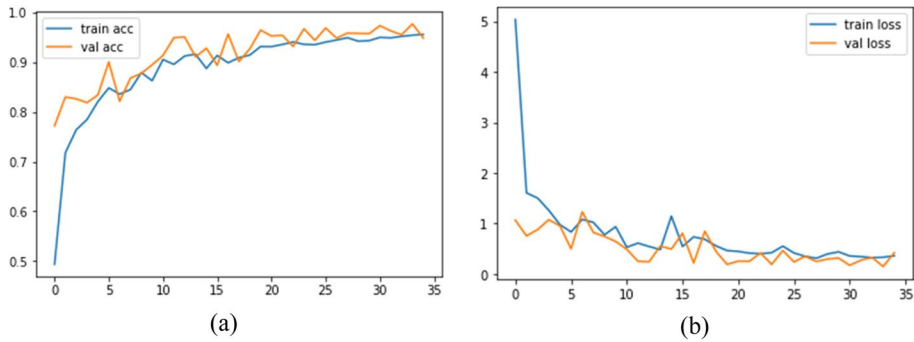


Fig. 28 Results plot of DenseNet121 with 35 epochs & 16 classes. X-axis represents the epoch and y-axis represents the accuracy in (a) and loss in (b)

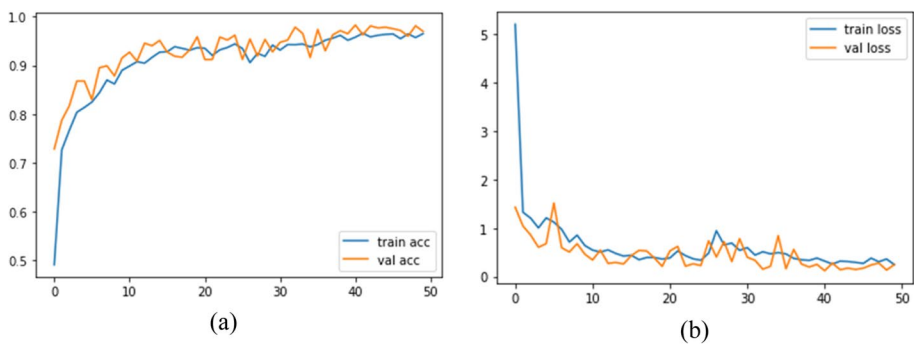


Fig. 29 Results plot of DenseNet121 with 50 epochs & 16 classes. X-axis represents the epoch and y-axis represents the accuracy in (a) and loss in (b)

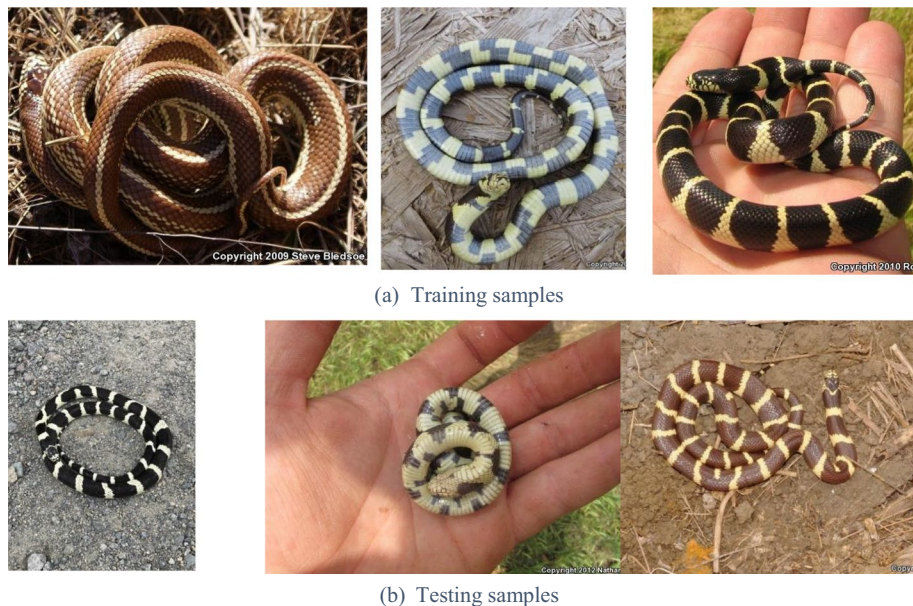


Fig. 30 class 448 samples

Table 8 CNN-based 20 snake species results

Model	# Of Epochs	Accuracy	Batch size	Time for each epoch
VGG16	35	97.65%	64	≈ 145 s
VGG16	50	97.93%	64	≈ 145 s
DenseNet121	50	97.11%	64	≈ 125 s
MobileNetV2	35	96.09%	64	≈ 85 s

The plot diagram when using 50 epochs and VGG16 architecture is shown in Fig. 32, plot between train accuracy and validation accuracy is shown in Fig. 32(a), plot between train loss and validation loss is shown in Fig. 32(b).

The plot diagram when using 50 epochs and DenseNet121 architecture is shown in Fig. 33, plot between train accuracy and validation accuracy is shown in Fig. 33(a), plot between train loss and validation loss is shown in Fig. 33(b).

The plot diagram when using 35 epochs and MobileNetV2 architecture is shown in Fig. 34, plot between train accuracy and validation accuracy is shown in Fig. 34(a), plot between train loss and validation loss is shown in Fig. 34(b).

4.2.5 CNN-based 22 snake species

The distinct CNN models are applied on 22 snake species, the class-id of the snakes that are included in this experiment are Class 67, Class 204, Class 337, Class 697, Class 771, Class 4, Class 72, Class 78, Class 128, Class 140, Class 239, Class 273, Class 326, Class 362, Class 390, Class 394, Class 448, Class 450, Class 460, Class 543, Class 581, and

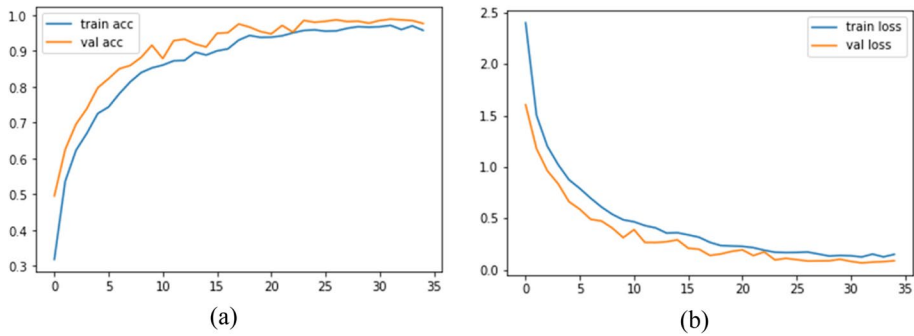


Fig. 31 Results plot of VGG16 with 35 epochs & 20 classes. X-axis represents the epoch and y-axis represents the accuracy in (a) and loss in (b)

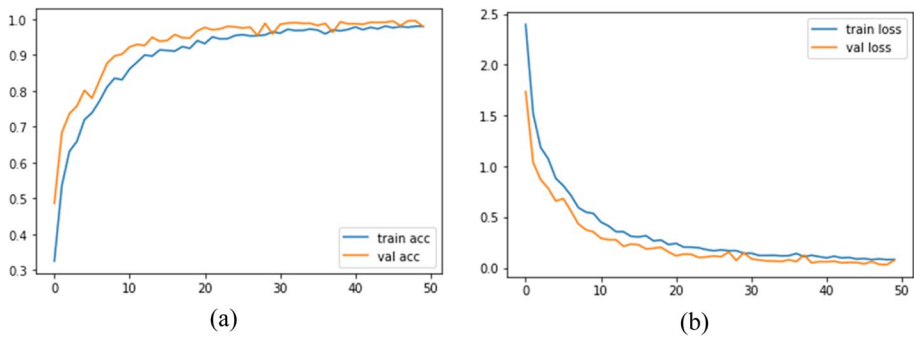


Fig. 32 Results plot of VGG16 with 50 epochs & 20 classes. X-axis represents the epoch and y-axis represents the accuracy in (a) and loss in (b)

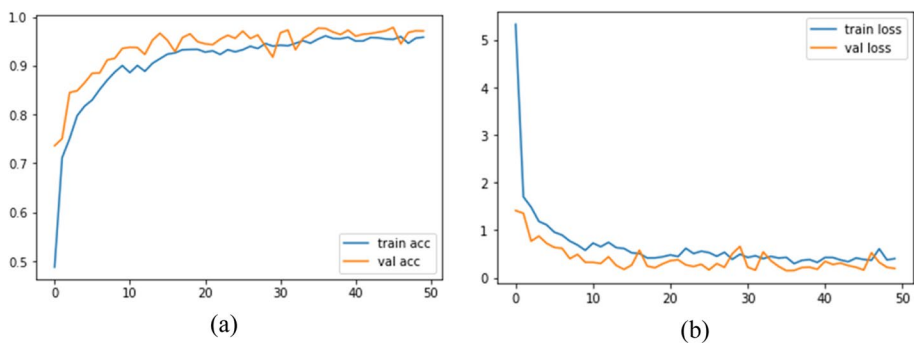


Fig. 33 Results plot of DenseNet121 with 50 epochs & 20 classes. X-axis represents the epoch and y-axis represents the accuracy in (a) and loss in (b)

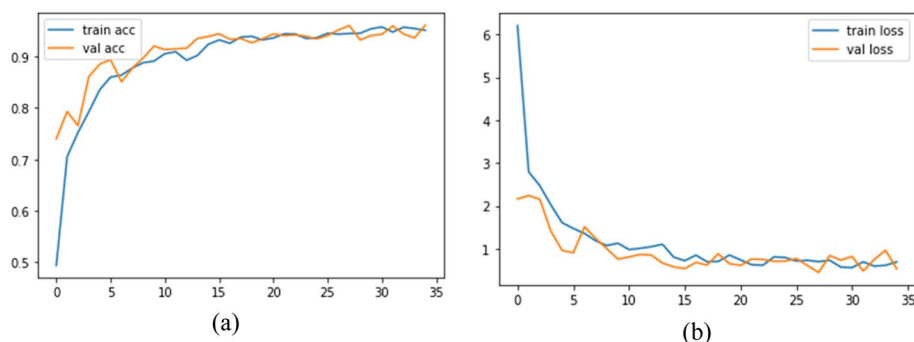


Fig. 34 Results plot of MobileNetV2 with 35 epochs & 20 classes. X-axis represents the epoch and y-axis represents the accuracy in (a) and loss in (b)

Class 707. Also, different epochs are used with batch size = 64. Total number of images = 7594, training images = 5289 images and testing images = 2305 images. Samples of training and testing images in class 581 are shown in Fig. 35.

All applied models are tested using data not seen before that was not involved in the training process.

The accuracy of CNN-based 22 snake species is shown in Table 9, and the plot diagram when using 35 epochs and VGG16 architecture is shown in Fig. 36, plot between train accuracy and validation accuracy is shown in Fig. 36(a), plot between train loss and validation loss is shown in Fig. 36(b).

The plot diagram when using 35 epochs and DenseNet121 architecture is shown in Fig. 37, plot between train accuracy and validation accuracy is shown in Fig. 50 (a), plot between train loss and validation loss is shown in Fig. 37(b).

The plot diagram when using 35 epochs and MobileNetV2 architecture is shown in Fig. 38, plot between train accuracy and validation accuracy is shown in Fig. 38(a), plot between train loss and validation loss is shown in Fig. 38(b).

4.2.6 CNN-based 45 snake species

The distinct CNN models are applied on all the dataset class which contains 45 snake species, the class-id of the snakes that are included in this experiment are Class 67, Class 204, Class 337, Class 697, Class 771, Class 4, Class 72, Class 78, Class 128, Class 140, Class 239, Class 273, Class 326, Class 362, Class 390, Class 394, Class 448, Class 450, Class 460, Class 543, Class 581, Class 707, Class 508, Class 526, Class 536, Class 540, Class 561, Class 597, Class 629, Class 635, Class 639, Class 653, Class 654, Class 663, Class 734, Class 784, Class 804, Class 811, Class 854, Class 857, Class 872, Class 957, Class 966, Class 1059, and Class 1625. Also, epochs = 35 are used with batch size = 64. Total number of images = 11285, training images = 7893 images and testing images = 3392 images. Samples of training and testing images in class 629 are shown in Fig. 39.

All applied models are tested using data not seen before that was not involved in the training process.

The accuracy of CNN-based 45 snake species is shown in Table 10, and the plot diagram when using 35 epochs and VGG16 architecture is shown in Fig. 40, plot between

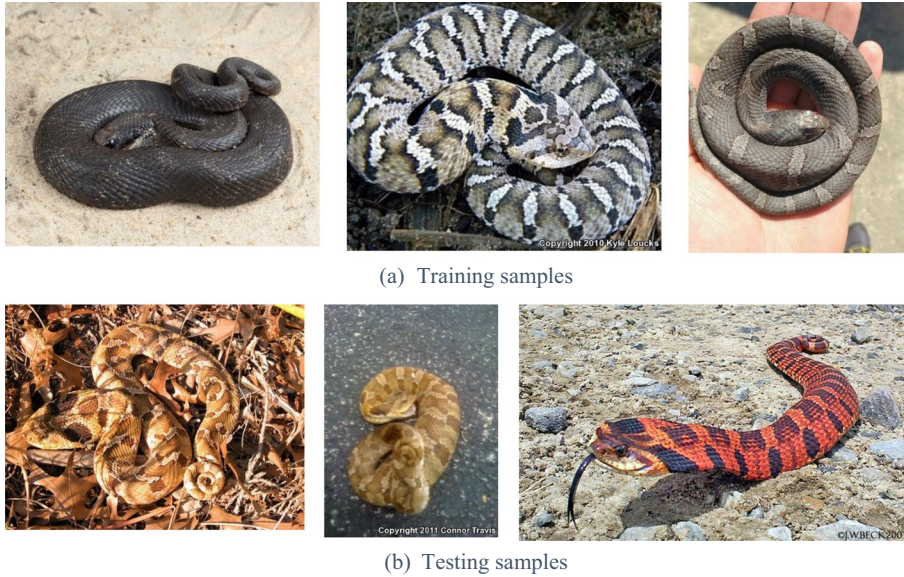


Fig. 35 class 581 samples

Table 9 CNN-based 22 snake species results

Model	# Of Epochs	Accuracy	Batch size	Time for each epoch
VGG16	35	98.01%	64	≈ 150 s
DenseNet121	35	96.33%	64	≈ 140 s
MobileNetV2	35	94.4%	64	≈ 100 s

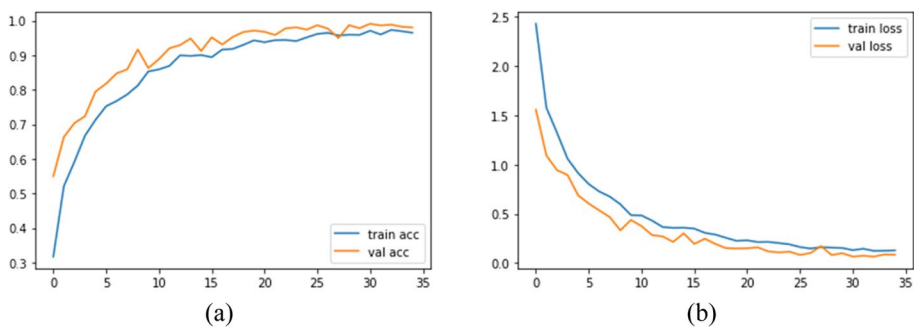


Fig. 36 Results plot of VGG16 with 35 epochs & 22 classes. X-axis represents the epoch and y-axis represents the accuracy in (a) and loss in (b)

train accuracy and validation accuracy is shown in Fig. 40(a), plot between train loss and validation loss is shown in Fig. 40(b).

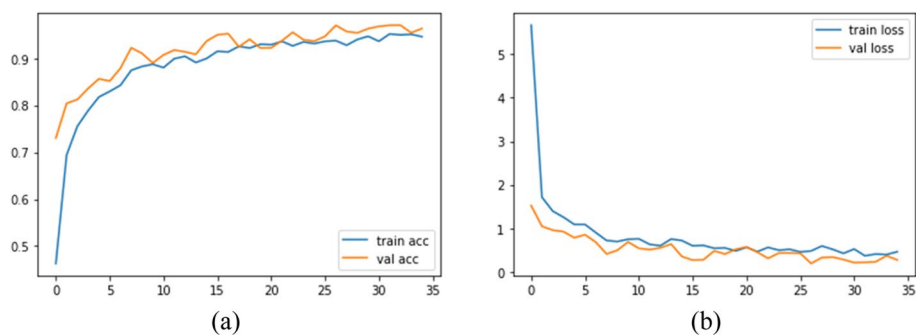


Fig. 37 Results plot of DenseNet121 with 35 epochs & 22 classes. X-axis represents the epoch and y-axis represents the accuracy in (a) and loss in (b)

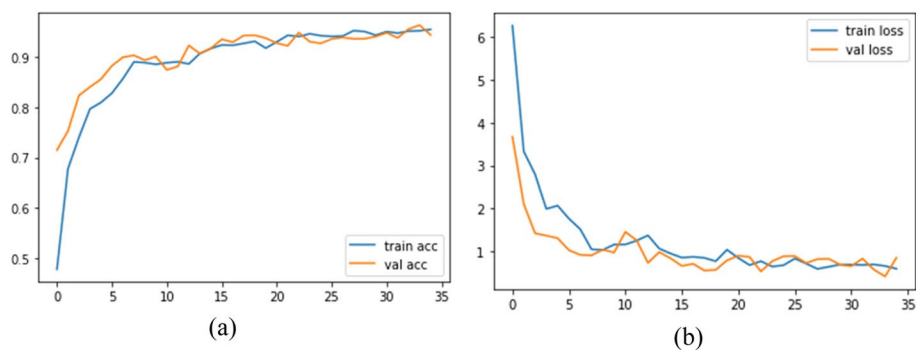


Fig. 38 Results plot of MobileNetV2 with 35 epochs & 22 classes. X-axis represents the epoch and y-axis represents the accuracy in (a) and loss in (b)

The plot diagram when using 35 epochs and DenseNet121 architecture is shown in Fig. 41, plot between train accuracy and validation accuracy is shown in Fig. 41(a), plot between train loss and validation loss is shown in Fig. 41(b).

The plot diagram when using 35 epochs and MobileNetV2 architecture is shown in Fig. 42, plot between train accuracy and validation accuracy is shown in Fig. 42(a), plot between train loss and validation loss is shown in Fig. 42(b).

4.3 Comparison with state-of-the-art

In this section a comparison between the proposed model and the state-of-the-art is presented (Tables 11 and 12).

Figure 43 illustrates the classes number used in each study of the state-of-the-art models.



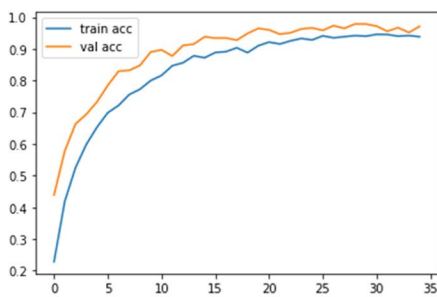
(a) Training samples



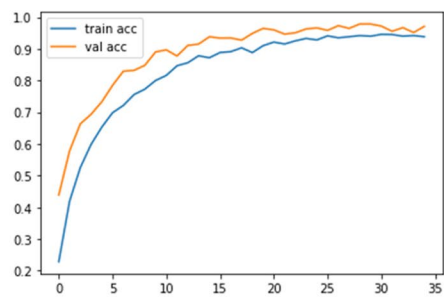
(b) Testing samples

Fig. 39 class 629 training samples**Table 10** CNN-based 45 snake species results

Model	# Of Epochs	Accuracy	Batch size	Time for each epoch
VGG16	35	97.09%	64	≈ 230
DenseNet121	35	95.34%	64	≈ 300
MobileNetV2	35	93.50%	64	≈ 140



(a)



(b)

Fig. 40 Results plot of VGG16 with 35 epochs & 45 classes. X-axis represents the epoch and y-axis represents the accuracy in (a) and loss in (b)

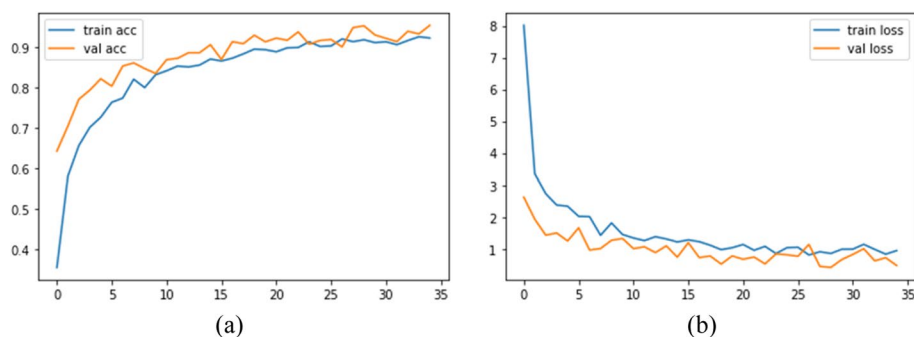


Fig. 41 Results plot of DenseNet121 with 35 epochs & 45 classes. X-axis represents the epoch and y-axis represents the accuracy in (a) and loss in (b)

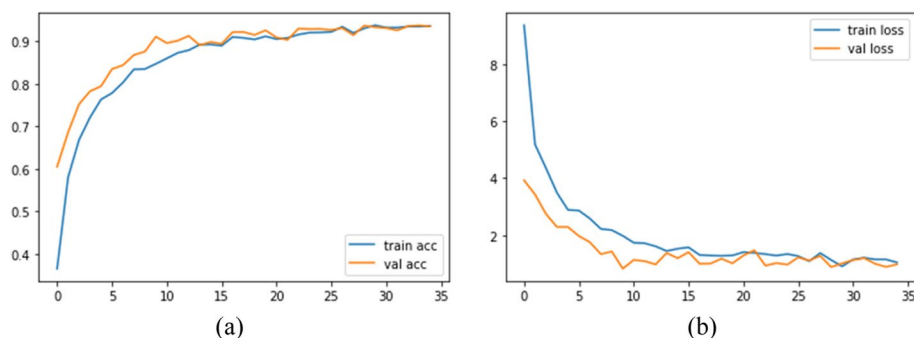


Fig. 42 Results plot of MobileNetV2 with 35 epochs & 45 classes. X-axis represents the epoch and y-axis represents the accuracy in (a) and loss in (b)

4.4 Discussion of results

The proposed model has achieved unprecedented results for classifying snake species. The model for SOD has detected the salient object (snake) accurately which led to improve the accuracy of the classification. There are 4 CNN architectures used in the classification process including VGG16, DenseNet121, MobileNetV2, and ResNet50. Softmax is used as an activation function which calculates the probability of the output as a value between 0 and 1, the target class is with the highest probability. Also, the Categorical Cross-Entropy loss function is used to judge the effectiveness of the network. The Adam optimizer is applied to update the weights of the network perfectly in an iterative manner in the training data. The results indicates that the ResNet50 with high number of parameter and achieved the lowest accuracy of 53.79% when applying to only 5 snake species while the VGG16 model is the best model for classifying snake species with accuracy 99.32%, 97.12%, 98.53%, 97.93%, 98.01%, 97.09% when using 5, 10, 16, 20, 22, and 45 number of classes, respectively. The accuracy of DenseNet121 comes in the second place. The accuracy was 98.92%, 95.54%, 96.79%, 97.11%, 96.33%, and 95.34% when using 5, 10, 16, 20, 22, and 45 snake species. While the MobileNetV2 achieved accuracy of 95.09%, 96.78%, 96.09%, 94.4%, and 93.5% when applying the model on 5, 10, 20, 22, and 45 snake species.

Table 11 comparison between our model and other tradition machine learning models

#	Research	Model	Year	Classes #	Images #	Accuracy
1	Ours	VGG16	2023	22	7594	98.01%
2		DenseNet121		22	7594	96.33%
3	[20]	LDA and SVM	2021	6	594	84%
4	[13]	NN	2017	22	349	89.22%
5		KNN		22	349	80.34%
6		Decision tree J48		22	349	71.29%
7		Naive Bayes		22	349	71.64%
8		Backpropagation NN		22	349	87.93%
9	[14]	KNN	2017	2	444	85.2%.
10	[15]	RBF	2014	6	1299	88.75%,
11		IBk		6	1299	87.50%
12		Multilayer perceptron		6	1299	86.64%

Table 12 comparison between our model and other tradition deep learning models

#	Research	Model	Year	Classes #	Images #	Accuracy
1		VGG16		45	11285	97.09%
2	Ours	DenseNet121	2023	45	11285	95.34%
3		VGG16		22	7594	98.01%
4		DenseNet121		22	7594	96.33%
5		VGG16		20	7245	97.93%
6		DenseNet121		20	7245	97.11%
7		VGG16		16	5775	98.53%
8		DenseNet121		16	5775	96.97%
9		VGG16		5	2354	99.32%
10	[26]	YOLOv5	2022	8	-	71%
11	[27]	Ensemble method	2022	2	1000	94.5%
12	[28]	Deep CNN	2021	2	1766	91.30%,
13	[39]	XResnet152	2021	45	82,601	87%
14	[20]	MobileNetV2	2021	6	594	93.16%
15	[21]	MobileNet	2020	9	247	10%
16		Inception V2		9	247	70%
17		VGG16		9	247	70%
18		ResNet		9	247	75%
19	[25]	DenseNet	2020	28	3050	72%
20	[16]	Medium CNN architecture	2019	5	415	82%
21	[17]	MobileNet	2019	6	2000	90.5%

In Fig. 44, it is observed that the proposed model gained about 10 points increase in performance compared to the used model in [39], this is due to the combination of using SOD algorithms with the Classification model which decreases the complex background

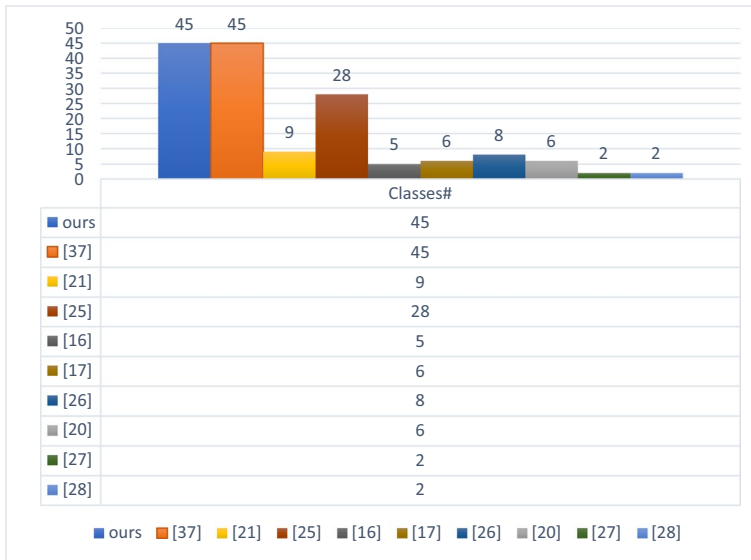


Fig. 43 number of classes in each study

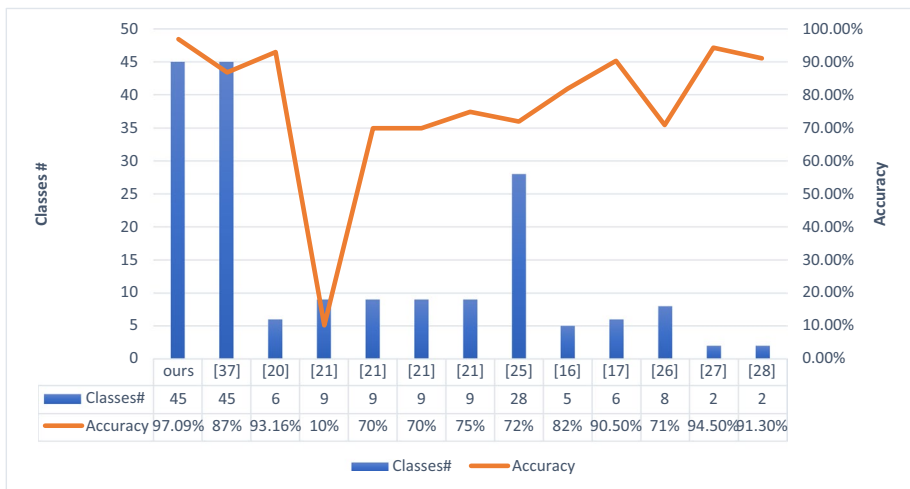


Fig. 44 Comparison between the classes and accuracy for the state-of-the-art

and extract the object to apply the classification algorithm to it directly that helps in accurately extract features and classify snakes.

As illustrated in Fig. 45, It is observed that The VGG16 is the model with best performance that achieved the highest accuracy compared to DenseNet121, MobileNetV2. In general, the model achieved high accuracy especially when using VGG16 in addition to applying SOD algorithm as a first step.

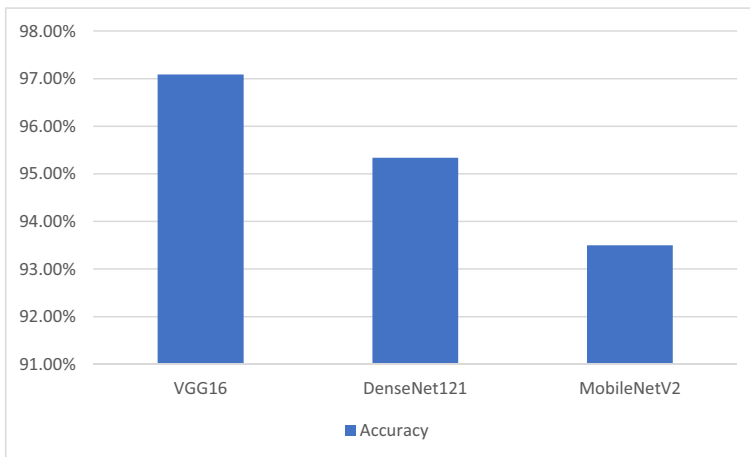


Fig. 45 Models Accuracy on 45 classes

5 Conclusion

The snake species classification has a significant role in drug discovery because it contains a wealth of bioactive proteins which is of great importance in this field, according to the latest research. Also know the snake species will help to save human life from death or any complications and disabilities. The complications include structural malformations, visual impairment, and kidney disease. When Snake species are visually recognized, this will lead to inaccurate classification and based on that in correct treatment will be given. The classification may be applied using traditional machine learning techniques which determine the features manually, the features may be heap shape, snake color, eye shape, body shape, and body texture. But not all people are familiar with the differences between snake species. So, the deep learning techniques is the best choice to classify snake species because in deep learning the features are set automatically and no need to have an experience in the difference between snakes. Also, deep learning techniques are dealing with large amounts of data and the snake datasets have large numbers of snake species and snake images. The proposed model helps to classify the snake species without the need for an expert in herpetology. The use of SOD greatly improved the results because of the bad effect of the complex background in classifying the snakes. When the salient object is detected before using a model for classification, this leads to a significant improvement in the results. As the results showed, the proposed model achieved unprecedented results. The SOD is applied to snake images using VGG16 architecture to detect the snake. Also applying data augmentation techniques to solve the overfitting problem which Significantly improved results. Then, 6 experiments are applied to different number of snake species using 4 CNN architectures to classify snake species. The proposed model is evaluated using snake images not seen before which are not included in the training phase. The VGG16 architecture achieved the best results in all conducted experiments, while the ResNet50 achieved the lowest accuracy. The comparison between the proposed model and other existing models showed that the proposed model has a higher performance than others with accuracy =97.09% when using 45 snake species.

6 Future work

The proposed model achieved high accuracy even if the snake species increase. But we are seeking to be always evolving so, in the future, when a GPU device with high specifications is available the number of classes used in the experiments will be increased. The proposed model will be applied to more snake species and the time spent will be significantly reduced. Also, an application with simple and friendly GUI will be developed to make it easier to deal with for people who don't have computer experience which will make the model being deployed more widely.

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Data availability The dataset is benchmark dataset from the ImageCLEF. It is available online at Aicrowd² website. The outdated version of this dataset is used which contains only 45 classes. The updated version of the dataset contains 414424 images for 772 snake species. All Python source code files for SOD, training and testing are available in the supplementary materials.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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² https://www.aicrowd.com/challenges/snakeclef2021-snake-species-identification-challenge/dataset_files

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